SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 3: User Testing, Security Checks & Iteration (LO5)

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1. Test Plan & Objectives (Lab 08)

The usability test evaluated whether farmers, buyers, and transporters could accomplish core workflows without assistance, measured error frequency, validated task completion times, and ensured zero security vulnerabilities exist in data exchange.

Test Parameters:

- Participant Count:** 8 participants (3 Farmers, 3 Wholesale Buyers, 2 Transporters).
- Format:** Moderated Usability Testing via Think-Aloud Protocol followed by the standardized System Usability Scale (SUS) questionnaire.
- Environment:** Physical field testing in Dambulla wholesale hub & simulated office environments in Colombo.

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2. Test Tasks & Completion Rates

Task ID	Task Description	Target User	Completion Rate	Avg Time (Sec)	Error Rate
T01	Register account & verify role	All	100%	34s	0%
T02	Add a new crop product with vi	Farmer	87.5%	78s	12.5%
T03	Browse marketplace, inspect ha	Buyer	100%	52s	0%
T04	Set up Branch Depot & request	Transporter	87.5%	68s	12.5%
T05	Confirm order pickup & update	Transporter	100%	41s	0%

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3. System Usability Scale (SUS) Results

The standardized 10-item SUS questionnaire was administered to all 8 participants.

SUS Score Calculation:

- Average SUS Score:** **84.5 / 100**
- Grade Equivalent:** **Grade A (Excellent Usability)**
- Adjective Rating:** Highly acceptable; exceeds the industry benchmark average of 68.0.

Key Qualitative Feedback:

- "I like that I can take a 10-second video of my tomatoes right from the tree.

Buyers immediately trust the quality."* ? Farmer, Dambulla.
- "The temperature tracking during delivery gives us complete confidence that leafy greens won't arrive spoiled."* ? Supermarket Buyer, Colombo.

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4. Secure Design & Data Privacy Checks

4.1 Authentication Flow Security:

- Enforced Firebase Phone Auth with reCAPTCHA verification to eliminate automated bot abuse.
- Biometric lock encryption enabled for financial wallet & escrow release screens.

4.2 Cloud Firestore Security Rules Verification:

- Role-Based Access Control (RBAC): Farmers can only mutate their own crop batches (‘request.auth.uid == resource.data.farmerId‘).
- Transporters can only modify assigned shipment states; financial escrow triggers require multi-signature approval.
- Verified that arbitrary document injections are rejected at the Firestore engine level.

4.3 Data Privacy & Media Storage:

- Sensitive proof-of-delivery photos and harvest videos are stored in authenticated storage buckets with expirable tokenized URLs.
- Sensitive payment tokens handled strictly via PCI-DSS compliant payment gateways (PayHere).

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5. Iterations & Improvements (Before vs. After)

Iteration 1: Empty Branch Depot Validation in Fleet Hub

- Observed Pain Point:** When transporters attempted to add a vehicle before creating a depot branch, the app threw a silent ‘StateError (No element)‘.
- Iteration Applied:** Added an early check ‘if (_branches.isEmpty)‘ that provides an immediate, informative snackbar message: *"Please add a Branch Depot first"* and guides them directly to branch registration.

Iteration 2: Harvest Video Auto-Deletion Upon Status Completion

- Observed Pain Point:** Farmers and system storage accumulated large video files for crops that were already sold and consumed.
- Iteration Applied:** Implemented an automated cleanup cycle where harvest videos uploaded for traceability are archived or removed from costly hot storage once the order reaches ‘Delivered & Escrow Released‘, preserving only the immutable batch QR audit hash.

Iteration 3: Low-Connectivity Field Caching

- Observed Pain Point:** Intermittent 4G cellular drops in remote rural farmlands caused form submission loss.
- Iteration Applied:** Enabled Firestore offline persistence (‘enablePersistence()‘) and client-side photo queueing, allowing farmers to

capture harvests offline with automatic background sync upon re-establishing connection.